

GUIDELINE NUMBER: GL-ICU-GEN-002
CATEGORY: ICU - GENERAL
DATE: June 2024
NEXT REVIEW DATE: June 2026

TITLE: Guideline for Care of the Patient Receiving Neuromuscular Blockade (NMB)

PURPOSE: Identify specific nursing interventions guidelines for patients receiving neuromuscular blocking agents. Neuromuscular blockade (NMB) is initiated to decrease metabolic energy expenditures and reduce impedance to mechanical ventilation by paralyzing the skeletal muscles.

APPLICABILITY/CRITERIA FOR USE:

- Adult critical care patients receiving neuromuscular blockade

SPECIAL INSTRUCTIONS:

I. Safety

- A. For patients on continuous NMB, and/or intermittent dosing of NMB, Train-of-Four (TOF) monitoring will be performed for the duration of NMB therapy at a minimum of every 1 hour or per provider order.
- B. For patients on continuous sedation with NMB, Bispectral Index Score (BIS) or EEG monitoring will be performed for the duration of the NMB/continuous sedation therapy at a minimum of every 2 hours or per provider order.
- C. BIS Sensor Pad Care: assess skin, change sensor pad, and alternate site every 24 hours.
- D. Ensure self-inflating resuscitation bag (ambu) with peep valve, face mask, dedicated airway low-constant suction, yankauer oral suction are at bedside and intubation equipment is readily accessible at all times
- E. Consider placing a sign at the HOB indicating that patient is receiving paralytics.
- F. Collaborate with provider & RRT for prescribed ventilator parameters upon initiation of paralytic
- G. Collaborate with physician regarding frequency of arterial blood gases (ABG)
- H. Evaluate effectiveness of ventilation and oxygenation

II. General Care

- A. Avoid verbal and tactical sensory overload
- B. Provide patient & family information related to regimen of neuromuscular blockade therapy
- C. Assess eyes with penlight for signs of corneal alterations (discoloration, drainage, redness, or irritation) every shift and collaborate with provider to initiate appropriate eye care. Consider eye care q4 hours and taping eyes if eyes can not remain closed.

III. Withdrawal of Paralytics

- A. During withdrawal/tapering of NMB, monitor for signs that indicate patient is emerging from paralytic:
 - 1. Muscle movement or twitching
 - 2. Spontaneous respiratory effort
 - 3. Alteration in vital signs
 - 4. Arterial blood gases
- B. Ensure that TOF equals 4/4 prior to withdrawal of ventilatory support and/or extubation.

IV. Documentation

- A. Document TOF and BIS scores when ordered with vital signs in the complex VS flowsheets
- B. Document NMB administration :
 - 1. Continuous infusions on the I&O flowsheet/MAR
 - 2. Document POC for safety precautions related to high risk for injury

KEYWORDS: neuromuscular blocking agent, paralytic, sedation, Train of Four (TOF), BIS, NMB

REFERENCES:

McHale- Wiegand, D. L. (2016). *AACN Procedure Manual for Critical Care: (7th.ed.)* St. Louis, Missouri : Elsevier Saunders.

SIGNATURES:

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